

Week 1: CVaR-MPPI for Highway Lane-Change Planning

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I. MAIN IDEA

I propose a risk-sensitive highway lane-change planner that extends the Model Predictive Path Integral (MPPI) control framework with a Conditional Value-at-Risk (CVaR) objective, replacing the standard expected-cost computation with a worst-case tail-risk Bellman backup to generate safer lane-change policies under uncertain surrounding traffic.

II. APPLICATION

Highway lane-changing is one of the most safety-critical maneuvers in autonomous driving, requiring real-time computation of dynamically feasible trajectories in the presence of uncertain, interactive surrounding vehicles. Production autonomy stacks at companies such as Waymo and Wayve employ sampling-based or optimization-based planners that must resolve this problem within strict latency budgets (≤ 100 ms). The dominant framework for sampling-based real-time planning is MPPI [1], which evaluates thousands of stochastic trajectory rollouts in parallel and fuses them via importance-weighted averaging. However, standard MPPI optimizes the expected cost over the rollout distribution. This is a risk-neutral objective that implicitly trades off rare but catastrophic outcomes (near-miss collisions) against frequent low-cost gains (faster lane-change completion). This is fundamentally misaligned with AV safety requirements, where the worst-case tail of the trajectory distribution (not its mean) determines whether a vehicle is safe. This project implements a CVaR-augmented MPPI planner in a simulated three-lane highway environment using a kinematic bicycle model, demonstrating that the proposed risk-sensitive planner produces measurably more conservative lane-change policies in near-miss traffic scenarios without sacrificing trajectory quality under normal conditions.

III. NOVELTY

Standard MPPI, as formalized in Williams et al. [1], minimizes a weighted average of trajectory costs, which is a risk-neutral approach. While effective for many driving tasks, this provides no guarantees on rare high-cost outcomes, which are exactly the scenarios that define AV safety limits. Conditional Value-at-Risk is a well-established risk measure introduced by Rockafellar and Uryasev [2] that computes the expected cost of the worst fraction of outcomes, making it well-suited for safety-critical planning. Chow and Ghavamzadeh [3] showed that CVaR can be optimized over Markov Decision Processes using a modified Bellman operator, providing the theoretical basis for

risk-sensitive dynamic programming. Nair et al. [4] first incorporated CVaR into MPPI but limited their evaluation to static obstacle environments and short-horizon tasks. Our work extends this to a multi-agent highway setting where surrounding vehicles follow stochastic motion models, and provides a direct comparison between standard MPPI and CVaR-MPPI across varying traffic densities and near-miss scenarios, which has not been studied in prior work.

IV. RELEVANCE TO DYNAMIC PROGRAMMING

This project is built on the stochastic optimal control problem, whose solution is described by the Hamilton-Jacobi-Bellman (HJB) equation, the continuous-time form of the Bellman optimality principle. Williams et al. [1] show that MPPI approximates the solution to this equation by representing the value function as an average over stochastic trajectory rollouts, where each rollout computes an approximate Bellman backup by summing stage costs forward in time. Extending this to CVaR requires replacing the standard Bellman expectation with the CVaR-Bellman operator derived by Chow and Ghavamzadeh [3], which tracks the expected cost in the worst quantile of outcomes rather than the mean. This is a direct and meaningful extension of dynamic programming to risk-sensitive decision making. The final report will derive the CVaR-augmented MPPI update rule from the discrete-time Bellman principle, keeping the dynamic programming connection at the core of the technical contribution.

V. USAGE OF AI

Claude (Anthropic) was used during the ideation phase to survey relevant literature and define the project scope. For the course project, Claude will be used to verify steps in the derivation of the CVaR-Bellman operator and to assist with debugging the Python simulation. All derivations will be cross-checked against the cited papers.

REFERENCES

- [1] G. Williams, A. Aldrich, and E. A. Theodorou, "Model predictive path integral control: From theory to parallel computation," *Journal of Guidance, Control, and Dynamics*, vol. 40, no. 2, pp. 344–357, 2017.
- [2] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.
- [3] Y. Chow and M. Ghavamzadeh, "Algorithms for CVaR optimization in MDPs," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 27. Curran Associates, Inc., 2014.
- [4] S. Nair, A. Balakrishnan, and N. Mehr, "CVaR-based safety in model predictive path integral control," in *Proceedings of the IEEE American Control Conference (ACC)*, 2022.